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Sequence-Based Investigation and Primer Design

Sequence: Arabidopsis thaliana

Project overview

This project focuses on the computational investigation of the CBF3 (DREB1A) gene from Arabidopsis thaliana. The coding sequence was identified using ApE, and a forward and reverse primer pair was selected. The primers were further evaluated using NCBI Primer-BLAST and IDT OligoAnalyzer to examine their sequence properties, specificity and potential secondary structures.

Aim

To retrieve the CBF3 (DREB1A) gene sequence of Arabidopsis thaliana from the NCBI database, identify its coding sequence (CDS), select a suitable forward and reverse primer pair using ApE software, and evaluate the primer properties and specificity using NCBI Primer-BLAST.

Gene selection rationale

The CBF3 gene, also known as DREB1A (Dehydration-responsive element-binding protein 1A), was selected from Arabidopsis thaliana.

CBF3 belongs to the AP2/ERF family of plant transcription factors and plays an important role in plant responses to cold stress. It binds to CRT/DRE cis-regulatory elements present in the promoters of stress-responsive genes and regulates their expression.

Biotechnological relevance:

- Helps understand the molecular mechanisms involved in plant cold-stress tolerance.
- Regulates the expression of genes associated with stress responses.
- Has potential applications in developing crops with improved tolerance to environmental stress.
- Provides a useful target for studying plant stress-response pathways.

Interpretation: CBF3 was selected because it is involved in an important plant stress-response pathway. Amplifying its coding sequence can help in studying the gene and preparing it for downstream molecular biology applications.

3. Database record retrieval.

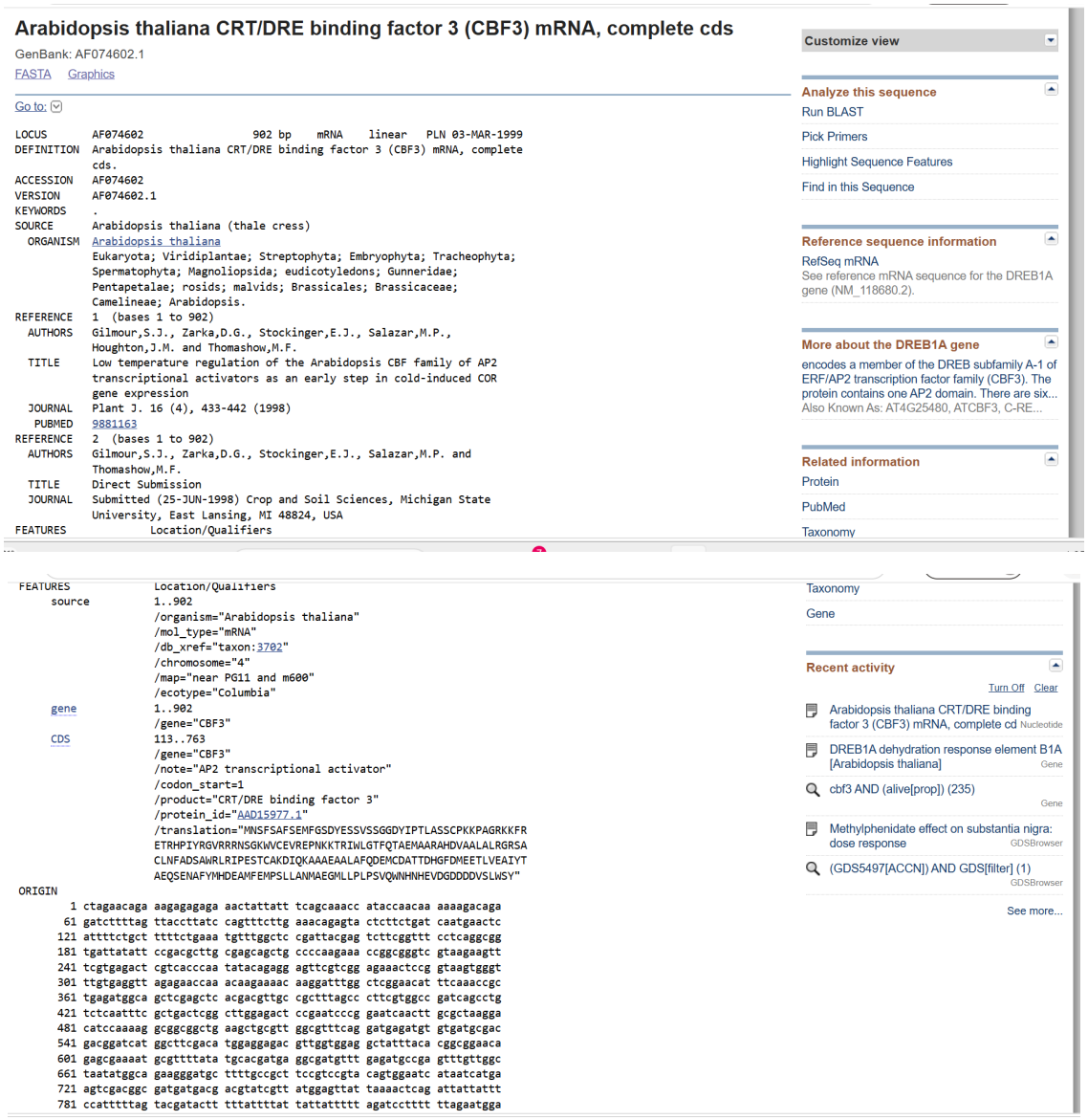


Figure:Result of DREB1A/CBF3 gene in Arabidopsis thaliana obtained from NCBI database

The GenBank record represents the CBF3 mRNA sequence of Arabidopsis thaliana. The accession number AF074602.1 uniquely identifies the sequence record and its version. The record contains a nucleotide sequence of 902 base pairs and is annotated with information about the gene and its coding region.

Parameter	**Details**
Full Name	Dehydration-responsive element-binding protein 1A
Common Alternative Name	C-repeat/dehydration-responsive element-binding factor 3 (CBF3)
Organism	Arabidopsis thaliana
Gene ID	828652
Locus ID	AT4G25480

Accession Number	AF074602.1
Sequence Length	902 bp
Molecule Type	mRNA
CDS Coordinates	113–763
CDS Length	651 bp
Chromosome	4
Protein Family	AP2/ERF transcription factor family

The CDS length calculation is:

$$763-113+1=651\text{bp}$$

The remaining nucleotides belong to the untranslated regions (UTRs).

Interpretation:

The selected record is appropriate for the intended task because it represents the CBF3 transcript and contains an annotated complete coding sequence. It provides a defined nucleotide template for selecting primers that amplify a region within the coding sequence.

The record is an mRNA sequence, not a genomic DNA sequence. Therefore, the sequence represents the processed transcript and does not contain the introns that may be present in the corresponding genomic gene.

Access date: 6 November 2026.

Identification of the coding sequence (CDS) and target region

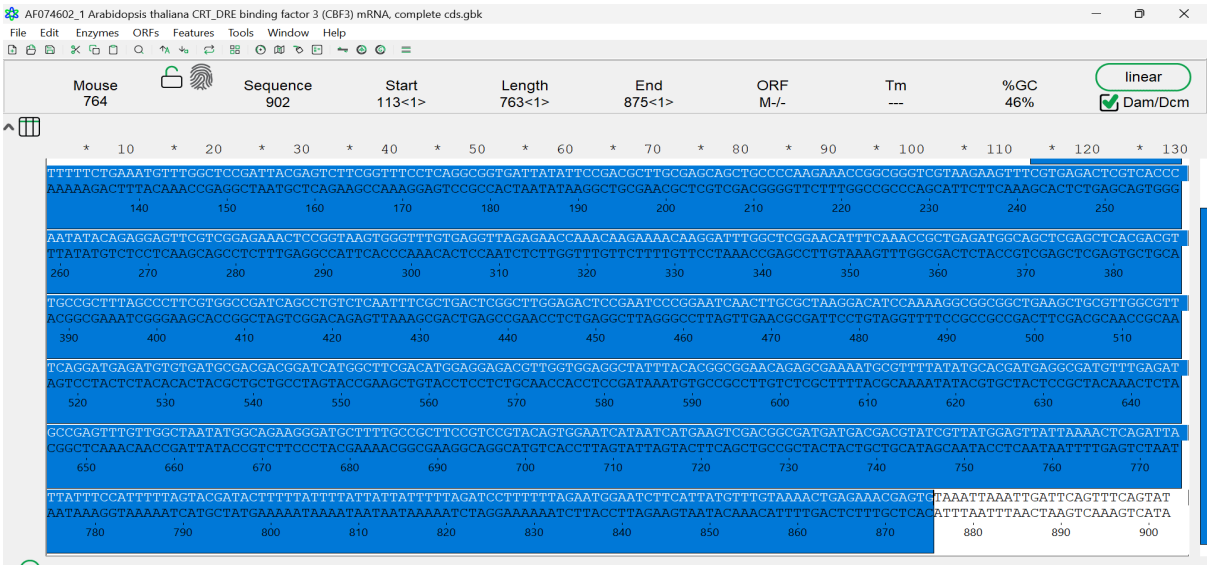


Figure: Identification of CDS region in ApE

The nucleotide sequence was opened in ApE, and the annotated coding region was examined. The CDS represents the portion of the mRNA that is translated into a protein. The selected primer-binding regions were located within the coding sequence.

Target-region selection:

The forward primer was selected from positions 156–176, while the reverse primer was selected from positions 693–716.

The selected regions lie within the annotated CDS (approximately positions 113–763, according to the supplied sequence information).

The expected amplified region is therefore located within the CBF3 coding sequence.

Interpretation:

Selecting primers within the CDS allows amplification of a defined portion of the protein-coding region. Since the template is mRNA-derived sequence, the corresponding experimental PCR template should be cDNA prepared through reverse transcription.

Primer design using ApE software

Software used: ApE (A Plasmid Editor)

ApE was used to examine the nucleotide sequence and select potential forward and reverse primers from the CBF3 coding region.

5.1 Selection of forward primer

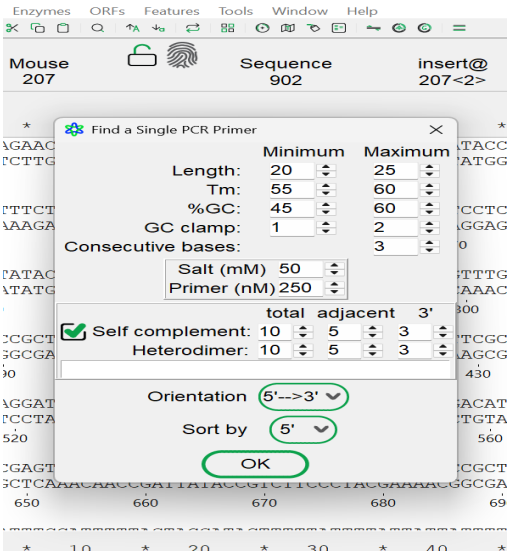


Figure: Parameters of selecting the forward primer

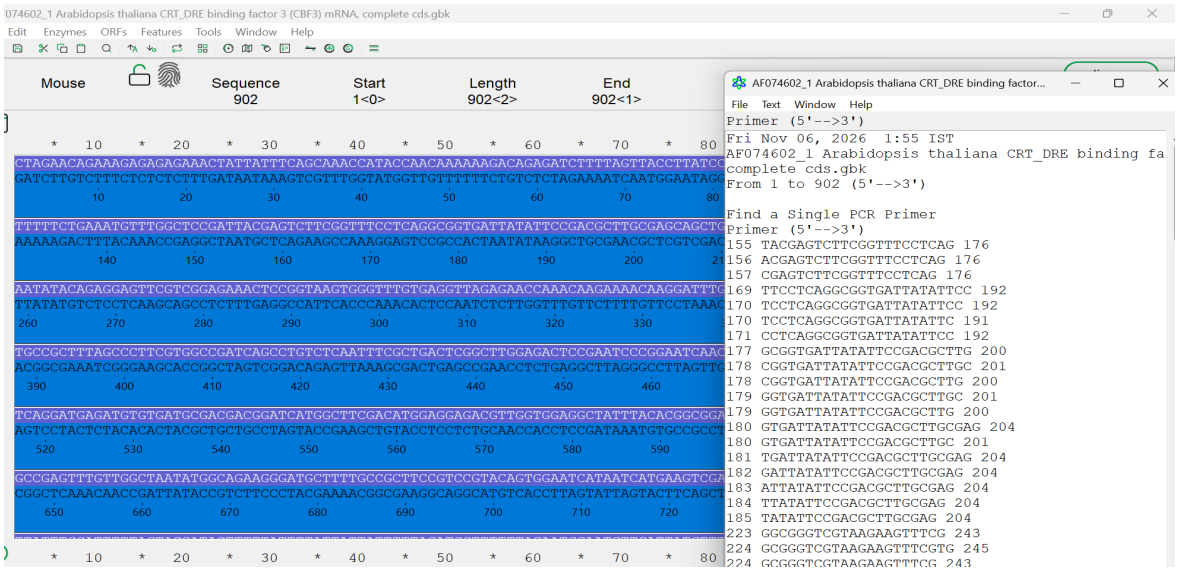


Figure: Result of Forward primers

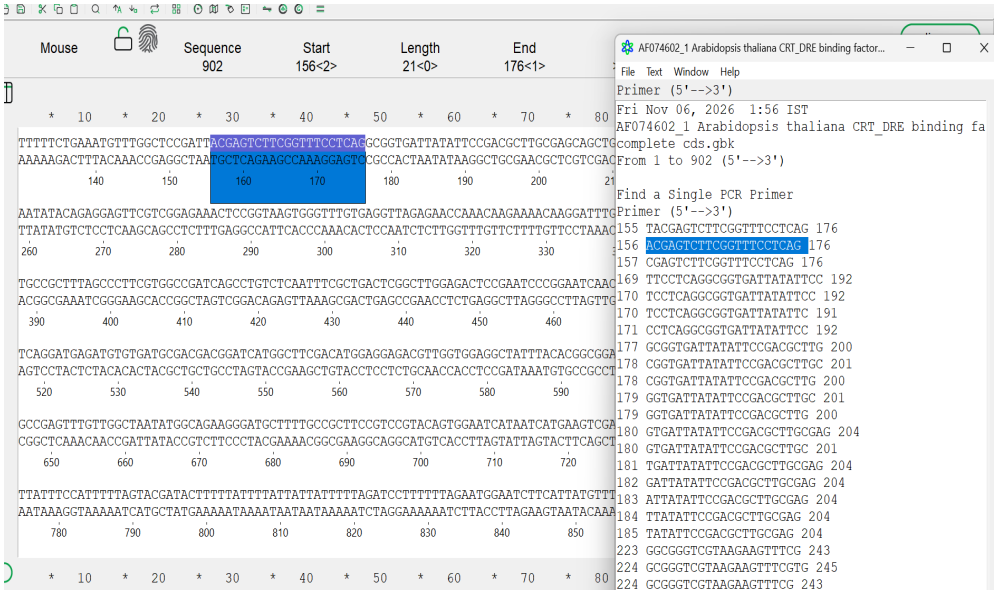


Figure: The selected forward primer

The forward primer was selected from the CBF3 mRNA sequence using the primer selection tool in ApE. The selected region was located between positions 156 and 176, with a length of 21 nucleotides.

Selected forward primer:

5′-ACGAGTCTTCGGTTTCCTCAG-3′

Interpretation:

The selected sequence acts as the forward primer and binds to the complementary strand of the template. It provides the starting point for DNA synthesis towards the target region.

5.2 Selection of reverse primer

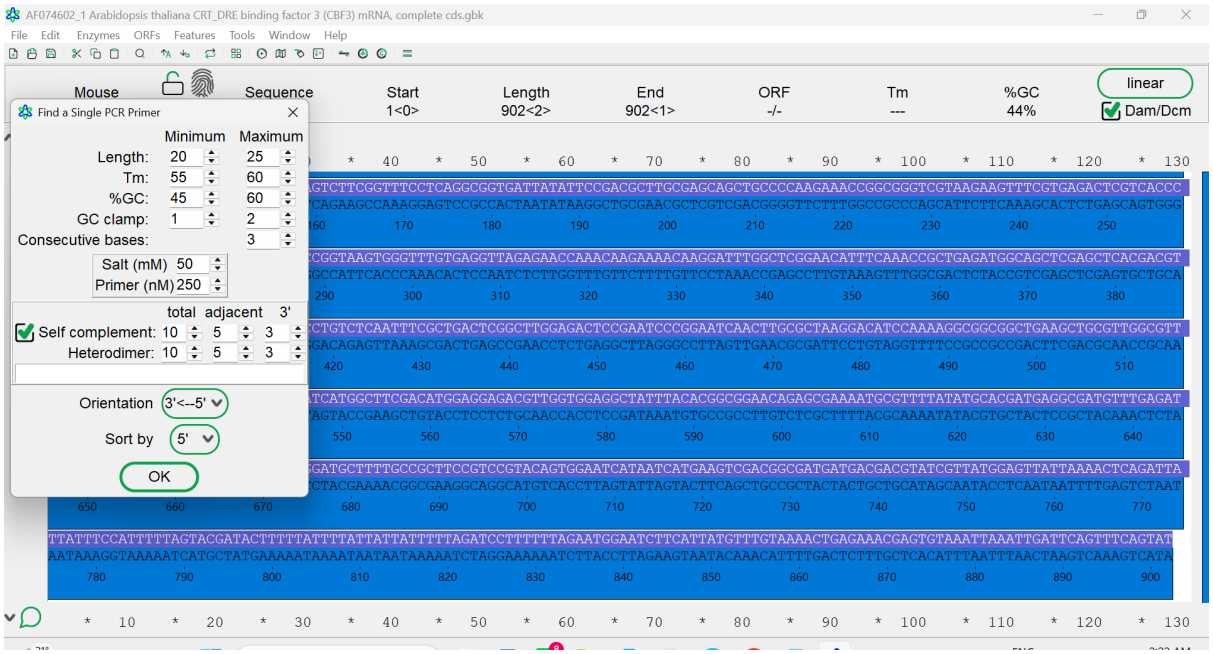


Figure: The selected parameters

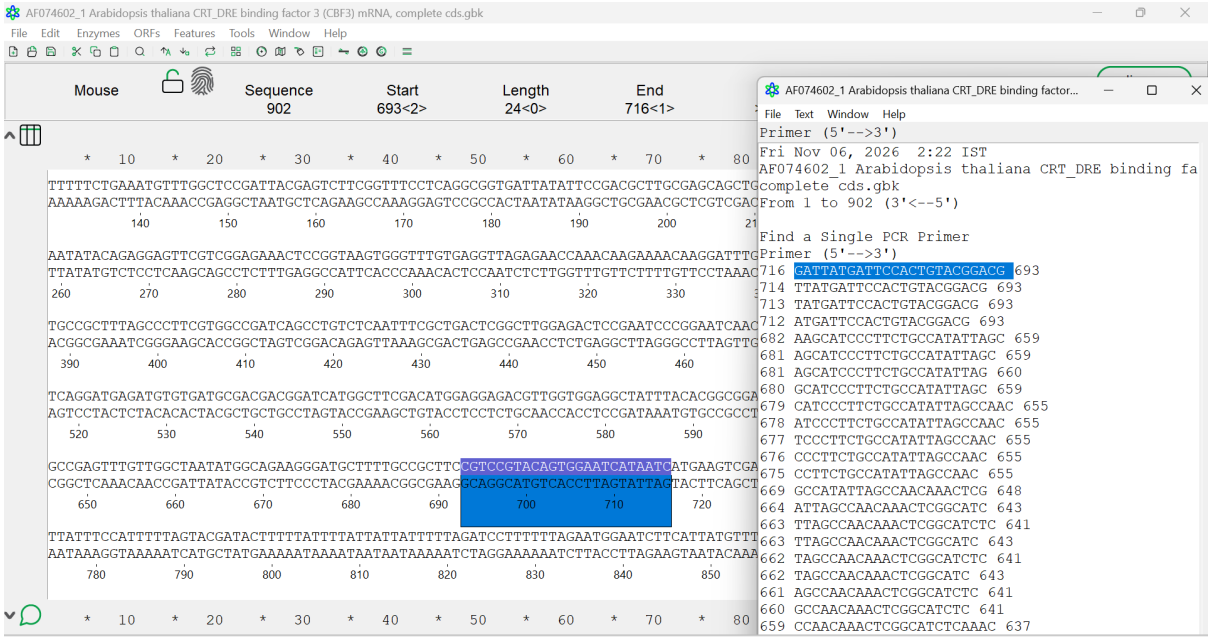


Figure: The selected Reverse Primer

reverse primer was selected from the region between positions 693 and 716 of the CBF3 sequence. The selected primer was 24 nucleotides long and was oriented on the opposite strand relative to the forward primer.

Selected reverse primer:

5′-GATTATGATTCCACTGTACGGACG-3′

Interpretation:

The reverse primer binds to the opposite strand and extends in the direction of the forward primer. The two primers face each other, allowing amplification of the intervening DNA region during PCR.

5.3 Summary of selected primers

Parameter	Forward primer	Reverse primer
Sequence (5′–3′)	ACGAGTCTTCGGTTTCCTCAG	GATTATGATTCCACTGTACGGACG
Position	156–176	693–716
Length	21 bp	24 bp
Strand	Plus (+)	Minus (−)

Expected amplicon size:

716-156+1=561bp

The predicted PCR product is therefore 561 bp.

Interpretation: The primers were selected from opposite strands and positioned to amplify a defined region of the CBF3 coding sequence.

6. Primer verification using NCBI Primer-BLAST

Tool used: NCBI Primer-BLAST

NCBI Primer-BLAST combines primer analysis with BLAST-based sequence searches. It helps evaluate whether a selected primer pair can amplify the intended target and whether similar binding sites may exist elsewhere in the selected database.

6.1 Entering the template and primer sequences

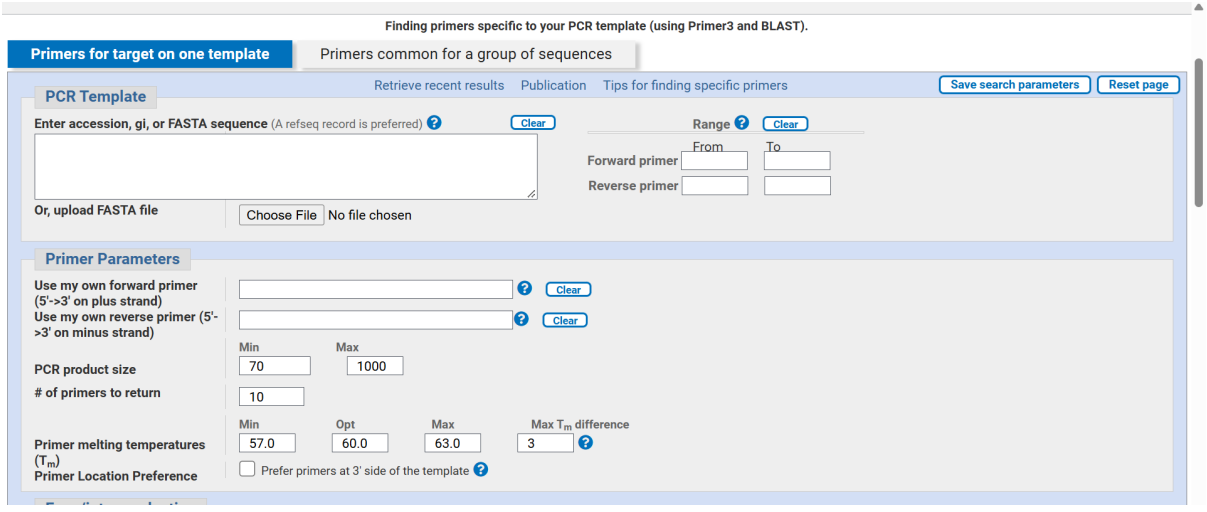


Figure: Homepage of NCBI-Primer BLAST

The CBF3 mRNA accession number was entered as the PCR template. The selected forward and reverse primers were entered in their respective fields, along with their binding positions. The organism was specified as Arabidopsis thaliana to restrict the specificity analysis to the selected organism.

Interpretation:

This step allowed the previously selected primers to be evaluated as a pair rather than as individual sequences. The specified binding positions helped the software identify the intended amplification region.

6.2 Graphical representation of primer binding

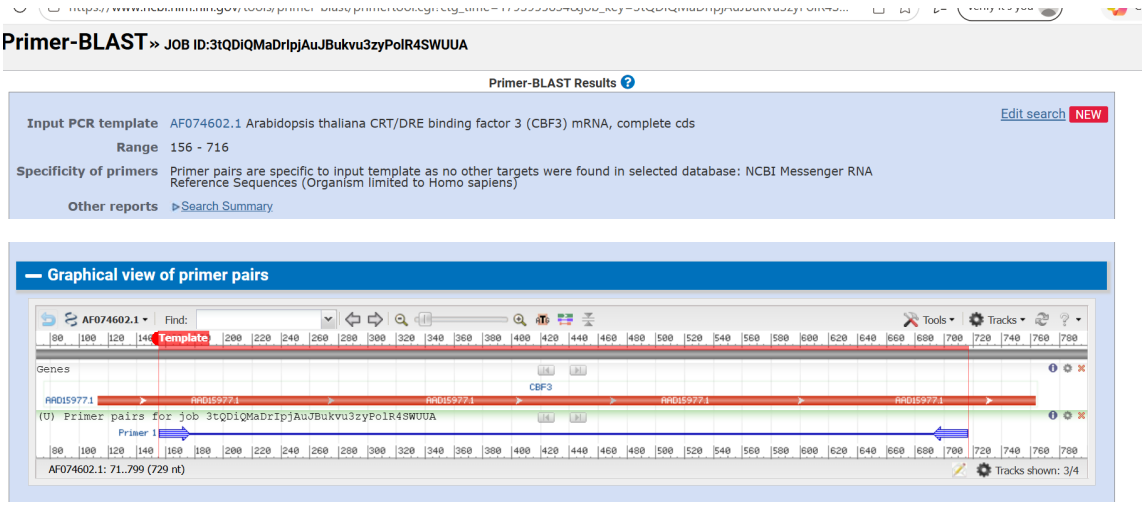


Figure : Graphical view of primer pair generated by Primer-BLAST

The graphical representation shows the selected primers' positions on the template sequence AF074602.1.

- The forward primer is represented on the plus strand.
- The reverse primer is represented on the minus strand.
- Both primers are oriented towards each other.
- The distance between the primer-binding regions corresponds to the predicted PCR product.

The graph also displays the gene annotation and the position of the CBF3 sequence within the template.

Interpretation:

The graphical view confirms that the selected primers bind to opposite strands and face each other. This arrangement is necessary for PCR amplification because DNA polymerase extends each primer in the 5′ to 3′ direction. The expected product includes the sequence between the two primer-binding sites.

6.3 Detailed primer report

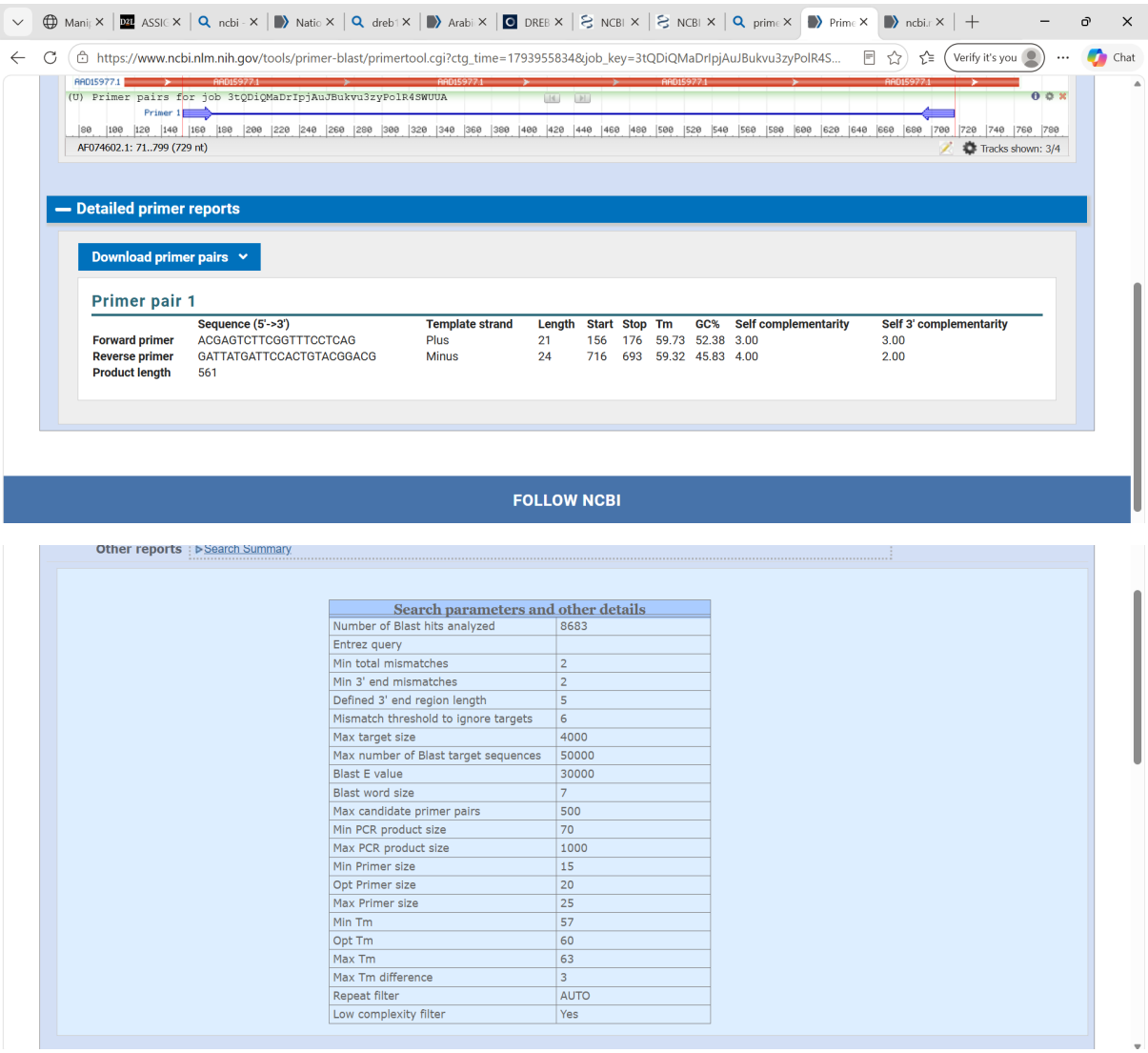


Figure: Detailed Primer Report

The detailed report provides the nucleotide sequences, strand orientation, primer length, binding positions, melting temperature, GC percentage and self-complementarity values for both primers.

The results obtained are presented below.

Parameter	**Forward primer**	**Reverse primer**
Sequence length	21 bp	24 bp
Start position	156	716
Stop position	176	693
Strand	Plus	Minus
Melting temperature	59.73°C	59.32°C
GC content	52.38%	45.83%

Self-complementarity	3	4
3′ self-complementarity	3	2

Note: Primer-BLAST displays the reverse primer's coordinates in descending order because it binds to the minus strand.

Interpretation of primer properties:

1. Primer length

The forward primer is 21 bp long, and the reverse primer is 24 bp long. Both lengths fall within the commonly used range for PCR primers.

2. Melting temperature (Tm)

The forward primer has a Tm of 59.73°C, while the reverse primer has a Tm of 59.32°C.

The difference is:

$$\Delta T_m = 59.73 - 59.32 = 0.41°C$$

The small difference indicates that both primers have similar melting temperatures, which is favourable for selecting a common PCR annealing temperature.

3. GC content

The forward primer has 52.38% GC content, and the reverse primer has 45.83% GC content.

Both values fall within the commonly accepted range of 40–60%. Suitable GC content helps maintain stable primer-template binding.

4. Self-complementarity

The forward and reverse primers have self-complementarity scores of 3 and 4, respectively. Their 3′ self-complementarity scores are 3 and 2.

Lower values are generally preferable because strong complementary interactions may result in primer-dimer formation. However, these scores alone do not establish whether secondary structures will form.

5. Product length

The predicted PCR product length is 561 bp. This represents the region between the forward and reverse primer-binding sites, including the primer sequences.

Overall interpretation:

The selected primer pair shows comparable melting temperatures, suitable GC content and low reported self-complementarity. These properties indicate that the primers are reasonable candidates for PCR amplification. Further secondary-structure analysis and experimental validation are required before confirming their performance.

6.4 Search parameters used in Primer-BLAST

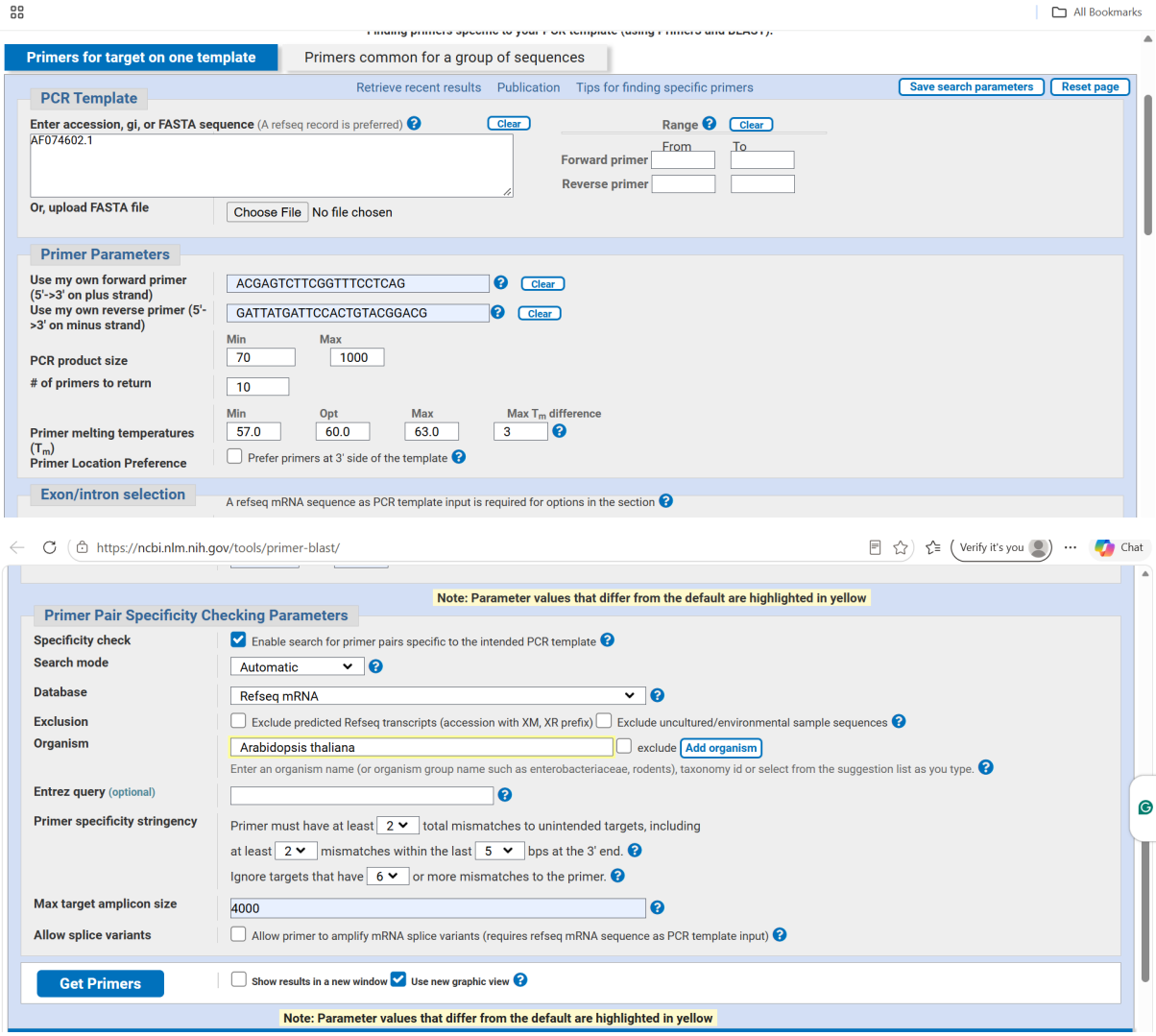


Figure : Input of primer sequences in NCBI Primer-BLAST

The search parameter table displays the conditions used by Primer-BLAST to evaluate the selected primer pair. The settings include the number of BLAST hits analysed, minimum total mismatches, 3′-end mismatch settings, PCR product size range, primer length limits and melting temperature limits.

The displayed settings include:

Parameter	**Value**
Minimum total mismatches	2
Minimum 3′-end mismatches	2
Defined 3′-end region length	5
Maximum target size	4000 bp
Minimum PCR product size	70 bp
Maximum PCR product size	1000 bp
Minimum primer size	15 bp
Optimum primer size	20 bp
Maximum primer size	25 bp
Minimum Tm	57°C
Optimum Tm	60°C

Maximum Tm	63°C
Maximum Tm difference	3°C

Interpretation:

The selected primers satisfy the displayed length and melting temperature ranges. The expected 561 bp product also falls within the specified PCR product size range of 70–1000 bp.

These settings determine how potential primer matches are evaluated. A specificity result is dependent on the database and search parameters used; it does not guarantee that nonspecific amplification will never occur under laboratory conditions.

7. Specificity analysis and limitations

7.1 Specificity of the selected primers

Primer specificity refers to the ability of a primer pair to amplify the intended target sequence without producing unwanted amplification products.

In Primer-BLAST, the selected primer pair was checked against the chosen sequence database. The results indicated that no additional targets were detected in the selected database.

Interpretation:

The result supports the possibility that the primer pair can specifically amplify the intended CBF3 sequence. However, specificity is dependent on the database searched, the organism selected and the mismatch criteria. It does not guarantee that the primers will produce only one band under actual laboratory conditions.

7.2 Possible primer secondary structures

Primer secondary structures can interfere with PCR amplification.

Three important structures should be checked using IDT OligoAnalyzer:

Structure	Explanation	Possible effect
Hairpin	A primer folds back and binds to a complementary region within itself.	Reduces availability for template binding.
Self-dimer	Two copies of the same primer bind to each other.	Competes with template binding.
Heterodimer	Forward and reverse primers bind to each other.	Can produce unwanted primer-dimer products.

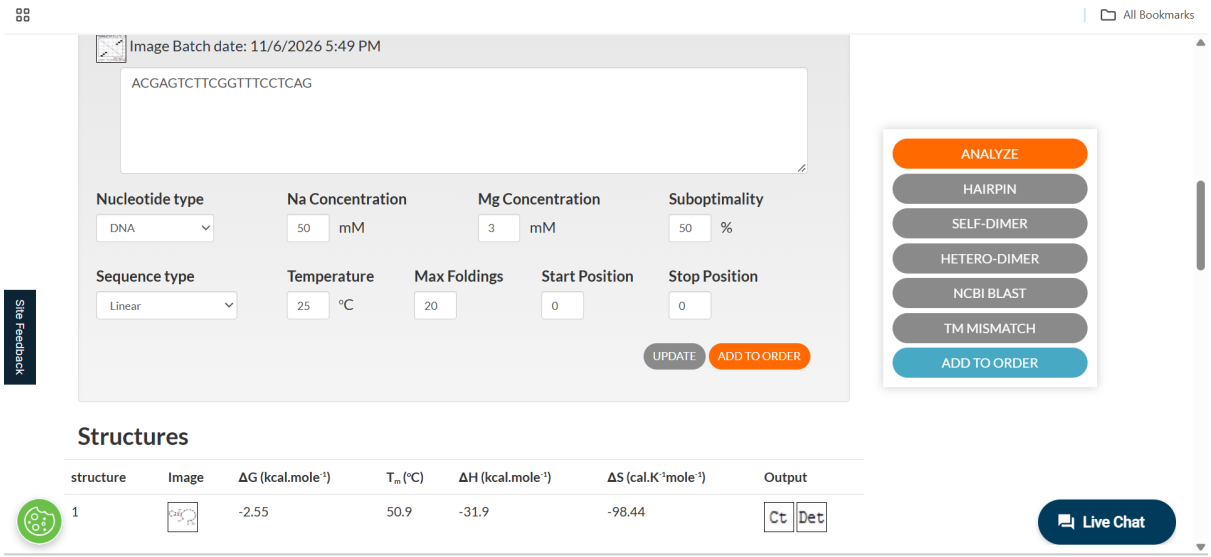


Figure: Hairpin Result for forward primer

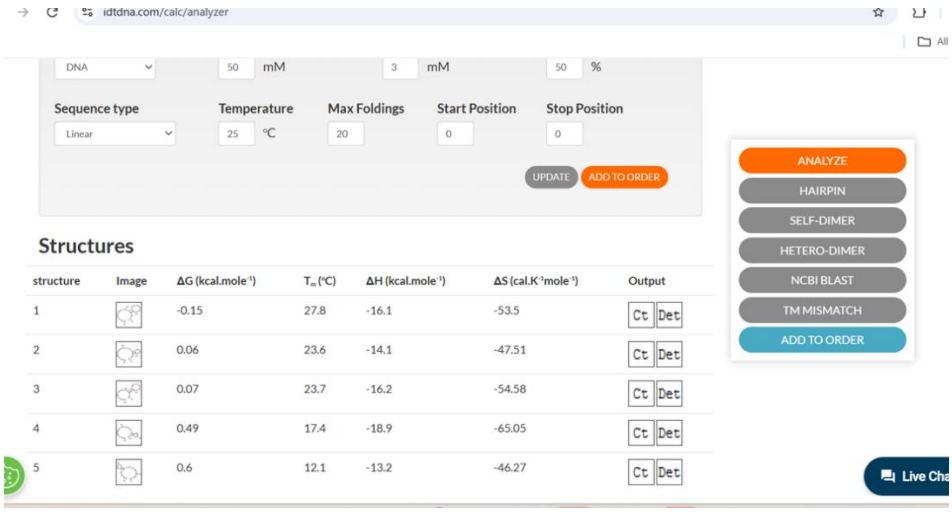


Figure: Hairpin result for reverse primer

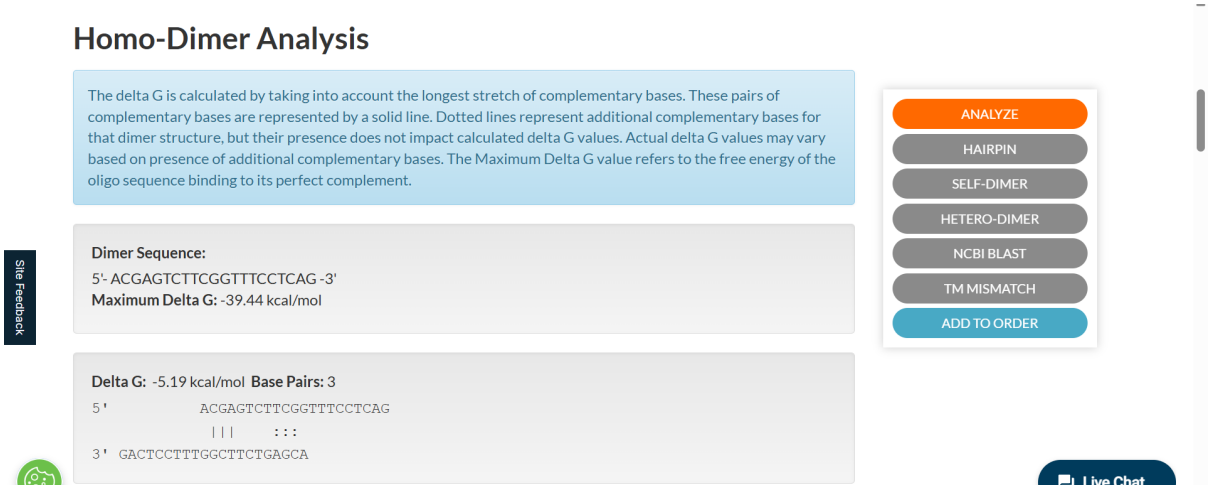


Figure: Self-Dimer result for forward primer

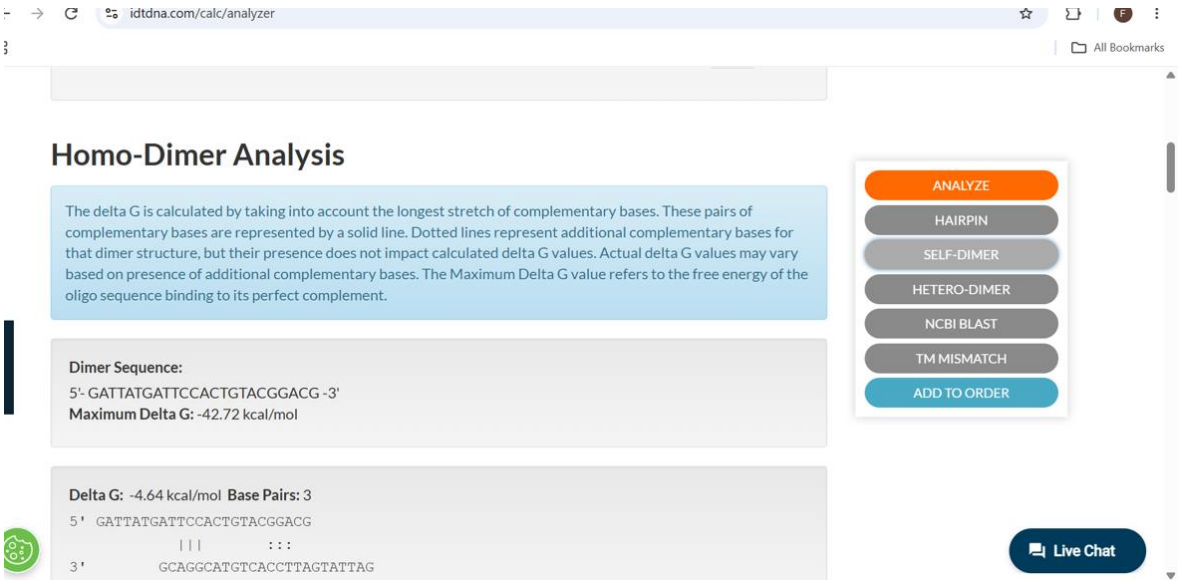


Figure: Self dimer result for Reverse primer

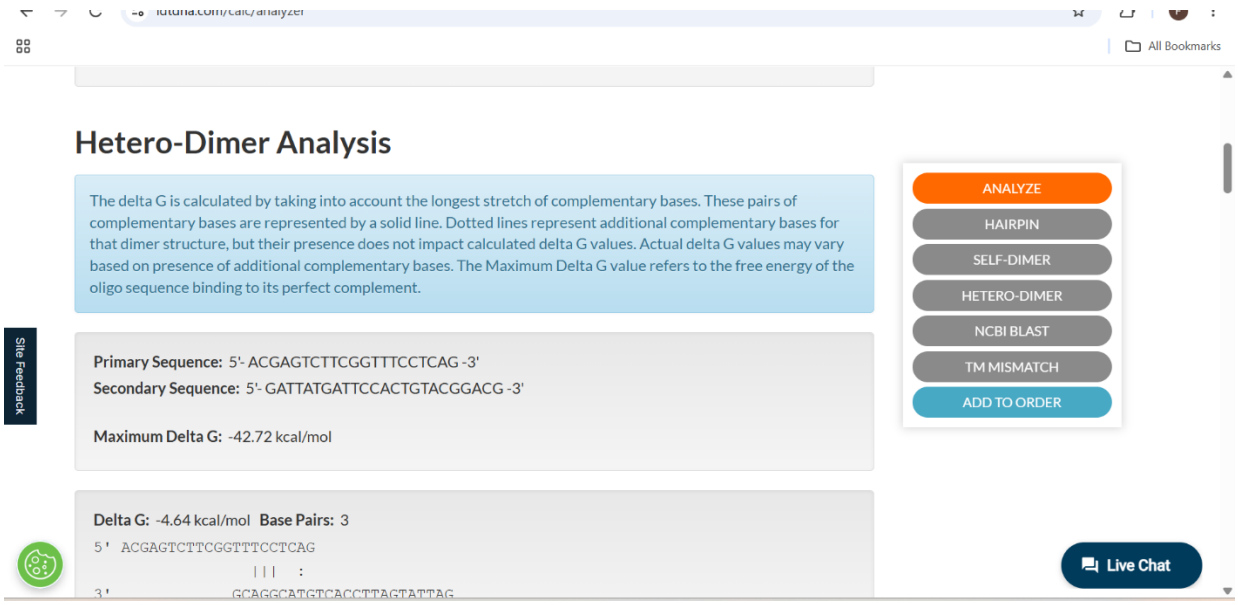


Figure: Hetero-Dimer Result

Interpretation of secondary-structure analysis

The secondary structures of the selected forward and reverse primers were evaluated using IDT OligoAnalyzer to examine the possibility of hairpin, self-dimer and heterodimer formation.

1. Hairpin formation

The forward primer showed a predicted hairpin melting temperature of 50.9°C and a ΔG of −2.55 kcal/mol. The relatively low free-energy value suggests that the predicted hairpin structure is not highly stable. This is favourable because strong hairpin formation can interfere with primer binding to the target DNA. The reverse primer's hairpin Tm is 27.8°C and ΔG is

−0.15 kcal/mol. These results suggest relatively weak hairpin formation for both primers, which is favourable for primer-template binding under typical PCR conditions.

2. Self-dimer formation

The forward primer showed the strongest displayed self-dimer interaction with a ΔG of −5.19 kcal/mol and three base pairs. The reverse primer showed the strongest displayed interaction with a ΔG of −4.64 kcal/mol and three base pairs.

Both results indicate relatively weak predicted self-dimer interactions, suggesting a low tendency for strong self-dimer formation.

3. Heterodimer formation

The interaction between the forward and reverse primers was evaluated using the heterodimer analysis option. The strongest displayed interaction had a ΔG of −4.64 kcal/mol and involved three base pairs.

This indicates a relatively weak interaction between the two primers, suggesting a low likelihood of significant heterodimer formation under the predicted conditions.

Overall interpretation

The secondary-structure analysis showed relatively weak predicted self-dimer and heterodimer interactions for the selected primer pair. The forward primer also showed a relatively weak predicted hairpin structure. These results are favourable for PCR primer selection because strong secondary structures can interfere with primer-template binding and amplification. However, experimental validation is still required to confirm primer performance under actual PCR conditions.

7.3 Limitations of primer design

Although the selected primer pair shows suitable initial properties, certain limitations should be considered:

1. Primer-BLAST provides computational predictions and does not confirm successful laboratory amplification.
2. Specificity depends on the selected database and organism settings.
3. Secondary structures need to be evaluated separately.
4. The selected mRNA sequence does not represent the complete genomic structure.
5. The predicted product length of 561 bp must be experimentally confirmed through PCR and agarose gel electrophoresis.
6. The selected region is part of the coding sequence, but amplification alone does not establish gene expression or protein function.

8. Intended use of the designed primers

The purpose of the selected primer pair is to amplify a 561 bp region within the CBF3 coding sequence of Arabidopsis thaliana.

This primer pair can be used for:

- Amplification of the selected CBF3 coding region.
- Preparation of a DNA fragment for downstream molecular biology applications.
- Further sequence analysis or cloning, subject to the requirements of the downstream experiment.

Template consideration:

The reference sequence used for primer design is mRNA. However, PCR requires a DNA template.

If the intention is to amplify the corresponding expressed transcript, RNA should first be isolated and converted into complementary DNA (cDNA) using reverse transcriptase.

If genomic DNA is used instead, introns may be present between the primer-binding regions. This can produce a larger PCR product or affect amplification, depending on the genomic structure.

In this experiment, the expected 561 bp product is based on the mRNA reference sequence. The corresponding product from genomic DNA should not automatically be assumed to have the same length.

9. PCR controls and expected results

After primer design and computational verification, the primer pair can be experimentally tested using PCR.

The following controls should be included:

Control	**Components**	**Purpose**
Positive control	Known CBF3-containing cDNA or suitable target DNA	Confirms that the PCR system can amplify the target.
Negative control	DNA/cDNA sample known to lack the target	Checks for unwanted amplification.
No-template control (NTC)	PCR mixture without DNA template	Detects contamination in PCR reagents or setup.

Expected gel electrophoresis result

Illustrative expected pattern: one band at approximately 561 bp in the positive control and test sample, with no target band in the negative control or NTC.

Interpretation of possible gel outcomes

Observation	Possible explanation	What to check
Single band at 561 bp	Expected amplification	Confirm band size using the DNA ladder.
No band	Poor template quality or unsuitable PCR conditions	Check template quality and annealing temperature.
Multiple bands	Nonspecific primer binding	Check primer specificity and optimize annealing temperature.
Small band near the bottom	Possible primer-dimer formation	Review heterodimer results and PCR conditions.
Band in NTC	Possible contamination	Repeat PCR using fresh reagents and clean setup.

Interpretation:

A single band at the expected size indicates successful amplification of a product consistent with the predicted target. However, gel size alone does not prove the identity of the product. Sequencing or another suitable verification method may be required to confirm that it is the intended CBF3 fragment.

Future scope

The selected primer pair can be further evaluated through experimental PCR using suitable cDNA as the template. The amplified product can be analysed by agarose gel electrophoresis and verified through sequencing to confirm the identity of the target fragment.

10. Conclusion

The CBF3 (DREB1A) mRNA sequence of Arabidopsis thaliana was retrieved from the NCBI GenBank database using accession number AF074602.1. The coding region was examined using ApE, and a forward primer of 21 bp and a reverse primer of 24 bp were selected.

The primer pair was evaluated using NCBI Primer-BLAST, which predicted an amplification product of 561 bp. The forward and reverse primers showed melting temperatures of 59.73°C and 59.32°C, respectively, and GC contents of 52.38% and 45.83%.

The initial results suggest that the selected primer pair is suitable for further evaluation. Secondary-structure analysis and experimental PCR validation are still required to establish its practical performance.

References

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